

Probabilistic Graphical Models CSE 5830

Course overview

Instructor

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Office Hours: Wednesday 11:00 AM - 12:00 PM, or by appointment

Course Description

Probabilistic Graphical Models (PGMs)

- Provide a powerful and flexible framework for reasoning and inference under uncertainty and for analyzing large-scale, complex, noisy, and heterogeneous data.
- Represent an intersection of probability theory, graph theory, statistics, and machine learning.
- Have wide-ranging applications in natural language processing, computer vision, healthcare and biomedical informatics, social and information networks, autonomous systems, and scientific discovery.

Course Description

This course consists of two parts:

- Conventional PGM
 - representation, including Bayesian and Markov networks
 - probabilistic inference algorithms, both exact and approximate
 - learning methods for both the parameters and the structure of graphical models
- Advance deep probabilistic and generative models
 - probabilistic Deep Learning and Bayesian Neural Networks
 - probabilistic Generative Models
 - diffusion and Score-Based Generative Models

Course topics

- | | | |
|-----------------------------------|---|---|
| PGM representation | { | <ul style="list-style-type: none">• Directed and undirected graphical model representation• Template-Based Graphical Models |
| PGM inference | { | <ul style="list-style-type: none">• Exact Inference by Variable Elimination• Belief Update and Message Passing• Approximate Inference as Optimization• Approximate Inference by Sampling• MAP Inference |
| Deep probabilistic and generative | { | <ul style="list-style-type: none">• Probabilistic Deep Learning and Bayesian Neural Networks• Variational Inference and Deep Latent-Variable Models• Probabilistic Generative Models: VAEs, Autoregressive Models• Diffusion and Score-Based Generative Models |
| PGM learning | { | <ul style="list-style-type: none">• Parameter Estimation in Directed and Undirected Graphical Models• Graphical Model Structure Learning |

Prerequisites

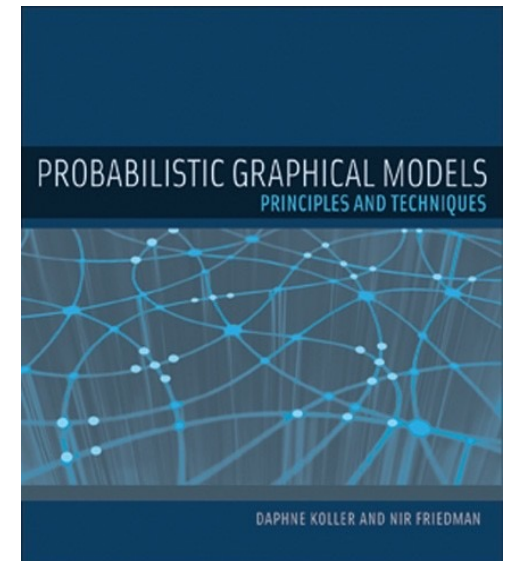
- Familiarity with programming
- Basic linear algebra (matrices, vectors, matrix-vector multiplication)
- Basic probability (random variables, distributions, densities, marginalization)
- Basic statistics (moments, typical distributions, regression)
- Algorithms (dynamic programming, basic data structures, complexity)

Texts and Materials

- Textbook

Probabilistic Graphical Models: Principles and Techniques, D. Koller & N. Friedman, MIT Press, 2009 (ISBN: 9780262013192).

<https://mitpress.mit.edu/books/probabilistic-graphical-models>

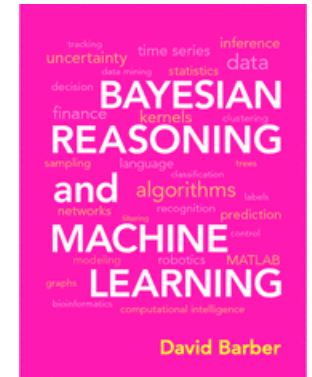


Texts and Materials

- Additional textbooks:

- *Bayesian Reasoning and Machine Learning*, David Barber, Cambridge University Press 2012. Free online.

<http://web4.cs.ucl.ac.uk/staff/D.Barber/pmwiki/pmwiki.php?n=Brml.HomePage>



- *An Introduction to Probabilistic Graphical Models*, Michael I. Jordan
Copies of certain chapters will be made available.

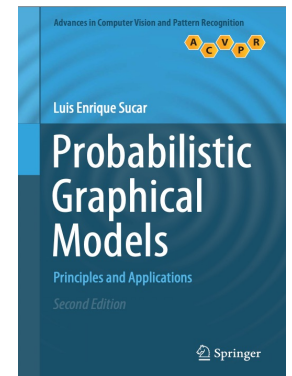
<https://www.cs.cmu.edu/~lebanon/pub/book/>

- *Probabilistic Graphical Models*, Luis Enrique Sucar, Springer, 2021

<https://link.springer.com/book/10.1007/978-3-030-61943-5>

- *Deep Generative Models*, Jakub M. Tomczak, Springer Nature, 2024.

<https://link.springer.com/book/10.1007/978-3-031-64087-2>



Outcome measure

- Class participation, pop quizzes
- Homework
 - There will be four assignments.
- Advanced topic presentation
 - Each student will present a recent paper, Second week of November
- Midterm
 - We will have a midterm exam Wednesday October 21st .
- Course Project.
 - Proposal
 - Presentation: Monday October 26th
 - Report: 2 pages, due Wednesday October 28th
 - Final project
 - Presentation: Last week of classes
 - Report: minimum 5 pages, In the format of IEEE conference papers, Due: Wednesday December 16th

Grading Criteria

Class participation	10%
Homework	10%
Midterm	20%
Proposal	5%
Advanced topics study	15%
Final report	35%
Final presentation	5%

Course Project

- There will be no final exam
- Final project
- The project must focus on a PGM or probabilistic deep learning problem.
- The proposal should clearly address what probabilistic model will be used and what inference problem will be solved.
- Projects may be theoretical or applied.

Project Proposal

Proposals should be approximately two pages long, and should include the following information:

- Project title and list of group members.
- Overview of project idea. This should be approximately half a page long.
- A short literature survey of 4 or more relevant papers. The literature review should take up approximately one page.
- Description of potential data sets to use for the experiments.
- Plan of activities, and how you plan to divide up the work.

Final Report

- ~5 pages
- In the format of the IEEE conference papers.
 - Introduction: problem definition and motivation
 - Background & Related Work: background info and literature survey
 - Methods: overview of your proposed method,
 - Results: Description of your experiments and results and a list of questions your experiments are designed to answer
 - Conclusion: discussion and future work

Advanced Topic Presentation

Second week of November

Format

- ~30-minute presentation
- 5-minute Q&A
- Slides required

Evaluation

- Assessed on: quality, clarity, depth of understanding
- Peer feedback will contribute to the final evaluation

Preparation

- Each student is required to select a paper from the provided list of papers on advanced topics in PGM
- Selecting paper is due on November 4th

Responsible Use of AI

- Use AI as a learning assistant to:
 - Clarify difficult concepts
 - Get feedback on your solution attempts
 - Ask for short hints when you are stuck
 - Explore or research new topics
- What AI Should NOT Do
 - Generate your final answers or solutions
 - Complete problem sets or assignments for you
 - Replace your own reasoning and problem-solving

Acknowledging use of AI

- Any use of AI for any problem set or other assignment must be properly documented and credited.
- For each such use, please include a brief (2-3 sentences) description of what specific tool was used, what materials are generated, and how that material was incorporated into the assignment.

Academic Integrity

We will follow the University Policy on Academic Integrity. For more information, see the related Student Conduct Rules. The URL of the UConn Student Conduct Rules web page is:

<http://community.uconn.edu/the-student-code-preamble/>

Accommodation

If you need any accommodation, please contact Center for Students with Disabilities (CSD), <https://csd.uconn.edu>.

